Ideal Fluids

Rather than just understanding the rate at which characteristics fluctuate at some fixed point, we consider how the momentum of a particle changes as it moves through space.

, since there are no local variations or changes in ū

A simple but powerful was to simplify the continuity equation is to consider the flow incompressible, $d\rho = 0$. Where for all all points in space and time density remains constant, $\rho(x, y, z, t) = C$. Thus satisfying $d\rho = 0$ and reducing the continuity equation to $\nabla \cdot \bar{u} = 0$.

The mass entering a system equals the mass exiting a system, mass is conserved in a closed volume. The total mass of a system ρV:

As this fluid is in motion, we understand it will remain until some force acts upon it due to the inertia of the fluid. Hence a particle in a moving fluid will remain in motion until some force acts on it. The net forces acting on this particle will be 0 if the velocity of the particle is constant. It is understood the acceleration of an object is directly proportional to the forces being applied and inversely proportional to the object's mass, otherwise stated as the sum of forces acting on an object is equivalent to the flux of momentum, $\Sigma F = ma$. When considering a single particle's motion the simple momentum equation is sufficient to describe the relationship between the momentum flux and the forces acting on the particle. Since the mass of the particle remains constant, for this reason any change in density through space and time must be 0, $\Delta\rho = 0$:

$$\Sigma F = m\frac{d\bar{u}}{dt}$$

$$\rho V = \iiint_V \rho dV$$

$$\oint \rho v \cdot df$$

$$\frac{\partial}{\partial t}\int \rho dV = -\oint \rho v \cdot df$$

$$\oint \rho v \cdot df = \int div(\rho v) dV$$

$$\int [\frac{\partial \rho}{\partial t} + div(\rho v)] dV = 0$$

$$\frac{\partial \rho}{\partial t} + div(\rho v) = 0$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho v_x}{\partial x} + \frac{\partial \rho v_y}{\partial y} + \frac{\partial \rho v_z}{\partial z} = 0$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} + \frac{\partial \rho v}{\partial y} + \frac{\partial \rho w}{\partial z} = 0$$

$$-\oint P df$$

$$-\oint P df = -\int \nabla P dV$$

$$\rho \frac{dv}{dt} = -\nabla P$$

$$dx\frac{\partial v}{\partial x} + dy\frac{\partial v}{\partial y} + dz\frac{\partial v}{\partial z} = (dr \cdot \nabla)v$$

$$dv = (\frac{\partial v}{\partial t})dt + (dr \cdot \nabla)v$$

$$\frac{dv}{dt} = \frac{\partial v}{\partial t}(v \cdot grad)v$$

$$\rho[\frac{\partial v}{\partial t}(v \cdot grad)v] = -gradP$$

$$\frac{\partial v}{\partial t}(v \cdot grad)v = \frac{1}{\rho}(-gradP)$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} = \frac{1}{\rho}(-\frac{\partial P}{\partial x})$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} = \frac{1}{\rho}(-\frac{\partial P}{\partial y})$$

$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} = \frac{1}{\rho}(-\frac{\partial P}{\partial z})$$

$$\Sigma F/V = m/V * a/V$$

$$\Sigma F/V = d/dt(m/V * a/V)$$

$$\Sigma F/V = d/dt(\rho * (u * u)/V)$$
$$\Sigma F/V = \rho(xyzt) * [(u(xyzt)/V(xyz))u(xyzt)]$$
$$\Sigma F/V = \rho(xyzt) * [u(xyz)/t + (u/x + u/y + u/z)u(xyz)]$$
$$\Sigma F/V = \rho(xyzt) * [\Delta\bar{u}/\Delta t + (\Delta\bar{u}/\Delta x + \Delta\bar{u}/\Delta y + \Delta\bar{u}/\Delta z)\bar{u}]$$
$$\Sigma F/V = \rho(xyzt) * [\partial\bar{u}/\partial t + (\partial\bar{u}/\partial x + \partial\bar{u}/\partial y + \partial\bar{u}/\partial z)\bar{u}]$$

$$\Sigma F/V = \rho[\partial u/\partial t + (\partial u/\partial x + \partial u/\partial y + \partial u/\partial z)\bar{u}] \quad \text{x-momentum}$$
$$\Sigma F/V = \rho[\partial v/\partial t + (\partial v/\partial x + \partial v/\partial y + \partial v/\partial z)\bar{u}] \quad \text{y-momentum}$$
$$\Sigma F/V = \rho[\partial w/\partial t + (\partial w/\partial x + \partial w/\partial y + \partial w/\partial z)\bar{u}] \quad \text{z-momentum}$$

$$\Sigma F/V = \rho[\partial u/\partial t + (\partial u^2/\partial x + \partial uv/\partial y + \partial uw/\partial z)] \quad \text{x-momentum}$$
$$\Sigma F/V = \rho[\partial v/\partial t + (\partial vu/\partial x + \partial v^2/\partial y + \partial vw/\partial z)] \quad \text{y-momentum}$$
$$\Sigma F/V = \rho[\partial w/\partial t + (\partial wu/\partial x + \partial wv/\partial y + \partial w^2/\partial z)] \quad \text{z-momentum}$$

$$\Sigma F/V = \rho[\partial\bar{u}/\partial t + (\nabla\cdot\bar{u})\bar{u}]$$

$$\Sigma F/V \rightarrow Pressure + Viscosity + Weight$$
$$\Sigma F/V \rightarrow P + \mu + mg$$

$$P + 1/2\rho v^2 + \rho gh = c$$
$$P + 1/2\rho v^2 = c$$
$$dP + \rho v dv = 0$$
$$\rho v dv =- dP$$
$$\rho[\partial\bar{u}/\partial t + (\nabla\cdot\bar{u})\bar{u}] =- \nabla P$$
$$- \nabla P =- [\partial P/\partial x, \partial P/\partial y, \partial P/\partial z]$$

$$[\sigma xx, \tau xy, \tau xz]$$
$$[\tau yx, \sigma yy, \tau yz]$$
$$[\tau zx, \tau zy, \sigma zz]$$
$$(\partial\sigma xx/\partial x + \partial\tau xy/\partial y + \partial\tau xz/\partial z)$$
$$(\partial\tau yx/\partial x + \partial\sigma yy/\partial y + \partial\tau yz/\partial z)$$
$$(\partial\tau zx/\partial x + \partial\tau zy/\partial y + \partial\sigma zz/\partial z)$$
$$\tau xy = \tau xy = \mu(\partial u/\partial y + \partial v/\partial x)$$
$$\tau xz = \tau zx = \mu(\partial u/\partial z + \partial w/\partial x)$$
$$\tau yz = \tau zy = \mu(\partial v/\partial z + \partial w/\partial y)$$
$$\sigma xx =- P + 2\mu(\partial u/\partial x)$$
$$\sigma xx =- P + 2\mu(\partial u/\partial x)$$
$$\sigma xx =- P + 2\mu(\partial u/\partial x)$$
$$\partial/\partial x(- P + 2\mu(\partial u/\partial x)) + \partial/\partial y(\mu * \partial u/\partial y + \mu * \partial v/\partial x) + \partial/\partial z(\mu * \partial w/\partial x + \mu * \partial u/\partial z)$$
$$- \partial P/\partial x + 2\mu(\partial^2 u/\partial x^2) + \mu(\partial^2 u/\partial y^2) + \mu(\partial/\partial y * \partial v/\partial x) + \mu(\partial/\partial z * \partial w/\partial x) + \mu(\partial^2 u/\partial z^2)$$
$$(- \partial P/\partial x)$$
$$\mu(\partial^2 u/\partial x^2) + \mu(\partial^2 u/\partial y^2) + \mu(\partial^2 u/\partial z^2) + \mu(\partial/\partial x * \partial u/\partial x) + \mu(\partial/\partial y * \partial v/\partial x) + \mu(\partial/\partial z * \partial w/\partial x)$$
$$\mu(\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2) + \mu * \partial/\partial x(\partial u/\partial x + \partial v/\partial y + \partial w/\partial z)$$
$$\mu(\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2) \quad \text{x-stress solution}$$
$$\partial/\partial y(- P + 2\mu(\partial v/\partial y)) + \partial/\partial x(\mu * \partial u/\partial y + \mu * \partial v/\partial x) + \partial/\partial z(\mu * \partial w/\partial y + \mu * \partial v/\partial z)$$
$$- \partial P/\partial y + 2\mu(\partial^2 v/\partial y^2) + \mu(\partial^2 v/\partial x^2) + \mu(\partial/\partial x * \partial u/\partial y) + \mu(\partial/\partial z * \partial w/\partial y) + \mu(\partial^2 v/\partial z^2)$$
$$(- \partial P/\partial y)$$
$$\mu(\partial^2 v/\partial y^2) + \mu(\partial^2 v/\partial x^2) + \mu(\partial^2 v/\partial z^2) + \mu(\partial/\partial y * \partial v/\partial y) + \mu(\partial/\partial x * \partial u/\partial y) + \mu(\partial/\partial z * \partial w/\partial y)$$
$$\mu(\partial^2 v/\partial x^2 + \partial^2 v/\partial y^2 + \partial^2 v/\partial z^2) + \mu * \partial/\partial y(\partial u/\partial x + \partial v/\partial y + \partial w/\partial z)$$
$$\mu(\partial^2 v/\partial x^2 + \partial^2 v/\partial y^2 + \partial^2 v/\partial z^2) \quad \text{y-stress solution}$$
$$\partial/\partial z(- P + 2\mu(\partial w/\partial z)) + \partial/\partial y(\mu * \partial w/\partial y + \mu * \partial v/\partial z) + \partial/\partial x(\mu * \partial w/\partial x + \mu * \partial u/\partial z)$$
$$- \partial P/\partial z + 2\mu(\partial^2 w/\partial z^2) + \mu(\partial^2 w/\partial y^2) + \mu(\partial/\partial y * \partial v/\partial z) + \mu(\partial/\partial x * \partial u/\partial z) + \mu(\partial^2 w/\partial x^2)$$
$$(- \partial P/\partial z)$$
$$\mu(\partial^2 w/\partial z^2) + \mu(\partial^2 w/\partial y^2) + \mu(\partial^2 w/\partial x^2) + \mu(\partial/\partial x * \partial u/\partial z) + \mu(\partial/\partial y * \partial v/\partial z) + \mu(\partial/\partial z * \partial w/\partial z)$$
$$\mu(\partial^2 w/\partial x^2 + \partial^2 w/\partial y^2 + \partial^2 w/\partial z^2) + \mu * \partial/\partial z(\partial u/\partial x + \partial v/\partial y + \partial w/\partial z)$$
$$\mu(\partial^2 w/\partial x^2 + \partial^2 w/\partial y^2 + \partial^2 w/\partial z^2) \quad \text{z-stress solution}$$
$$\mu\nabla^2\bar{u}$$

$$w = mg$$
$$w/V = m/V * g/V$$
$$w/V = \rho g(xyz)$$

$$\Sigma F/V \rightarrow \; - \nabla P + \mu \nabla^2 \bar{u} + \rho g$$

$$\rho[\partial \bar{u}/\partial t + (\nabla \cdot \bar{u})\bar{u}] = - \nabla P + \mu \nabla^2 \bar{u} + \rho g$$

x-momentum

$$\rho[\partial u/\partial t + (\partial u^2/\partial x + \partial uv/\partial y + \partial uw/\partial z)] = - \partial P/\partial x + \mu(\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2) + \rho g$$

y-momentum

$$\rho[\partial v/\partial t + (\partial vu/\partial x + \partial v^2/\partial y + \partial vw/\partial z)] = - \partial P/\partial y + \mu(\partial^2 v/\partial x^2 + \partial^2 v/\partial y^2 + \partial^2 v/\partial z^2) + \rho g$$

z-momentum

$$\rho[\partial w/\partial t + (\partial wu/\partial x + \partial wv/\partial y + \partial w^2/\partial z)] = - \partial P/\partial z + \mu(\partial^2 w/\partial x^2 + \partial^2 w/\partial y^2 + \partial^2 w/\partial z^2) + \rho g$$

$$W = \frac{1}{2} m u^2{}_{end} - \frac{1}{2} m u^2{}_{initial} = \Delta(\frac{1}{2} m u^2).$$

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Heat Equation Solutions

Let's derive a solution for the one dimensional Heat Equation. We have a partial differential equation that determines how heat will flow through a system. For this particular equation itself as we had previously stated temperature (T) as a function of time as well as a function of space and α being the thermal diffusivity. In a three dimensional form T is a function of (x,y,z,t) therefore for its one dimensional form we can conclude T(x,t). We arrive to the following equation:

$$\frac{\partial T}{\partial t} = \alpha(\frac{\partial^2 T}{\partial x^2})$$

We can form a numerical solution for this equation using the finite difference method. The finite difference method approximates derivatives with manageable differences. Let's look at the limit of the first and second derivatives to begin setting up the equation. Starting with the first derivative:

$$\lim_{dt \to 0} dT/dt = (T(t + dt) - T(t))/dt$$

If the limit of the first derivative as stated above finds what happens when the difference dt approaches 0. It must be stated that this is a forward difference where you are taking the difference of t as it increases by dt therefore we are determining the difference of each increasing time step. Rather than find the limit as dt becomes an infinitely small number we can assume the dt is definable and finitely small. For example we can have dt = 0.1. We can then formulate the following approximation:

$$dT/dt \approx (T(t + dt) - T(t))/dt$$

This is the finite difference for the first derivative. It can closely approximate the first derivative of the temperature function with respect to time. Now lets evaluate the limit of the second derivative of the original equation:

$$\lim_{dx \to 0} dT/dx = (T(x + dx) - 2T(x) + T(x - dx))/dx^2$$

The limit we have formed is the central difference. This limit allows us to figure out a finite difference for the second derivative in a similar manner as to our previous approximation. We can set dx to definable and finitely small number and set up our approximation:

$$dT/dx \approx (T(x + dx) - 2T(x) + T(x - dx))/dx^2$$

Now that we have formed our two approximations for our first and second derivative we can plug them back into the original heat equation. What should result will be the following equation:

$$(T(t + dt) - T(t))/dt = \alpha[(T(x + dx) - 2T(x) + T(x - dx))/dx^2]$$

With our new equation we need to define variables and boundaries. Our boundaries define a set area and initial values in which our simulation will run. Since we are examining a one dimensional heat transfer, we only need to worry about a one dimensional spatial area defined by our length of the x-axis. We know each x-value has a temperature value to represent it:

$[x_1 \to T_1]$
$[x_2 \to T_2]$
$[x_3 \to T_3]$
$[x_n \to T_n]$

If each x-value represents a certain temperature and we know each temperature value is dependent on the temperature of the x-values around it. It's also understood temperature will change based on each time step forward. So let's develop an understanding the relationship between our temperature and time:

$[T_1 \to T_1(t_1)T_1(t_2)T_1(t_3)\ldots T_1(t_m)]$
$[T_2 \to T_2(t_1)T_2(t_2)T_2(t_3)\ldots T_2(t_m)]$
$[T_3 \to T_3(t_1)T_3(t_2)T_3(t_3)\ldots T_3(t_m)]$

$[T_n \rightarrow T_n(t_1)T_n(t_2)T_n(t_3)\ldots T_n(t_m)]$

Lets define a new matrix that represents temperatures relation with time and space. We can let (m) represent all time steps (dt) and (n) represents all increments of x-values (dx):

$[T^1{}_1 T^2{}_1 T^3{}_1 T^m{}_1]$
$[T^1{}_2 T^2{}_2 T^3{}_2 T^m{}_2]$
$[T^1{}_3 T^2{}_3 T^3{}_3 T^m{}_3]$
$[T^1{}_n T^2{}_n T^3{}_n T^m{}_n]$

Both time and space have boundary conditions as well as their increments of change. Let's define our spatial boundaries. Our one dimensional space is represented by x which is a finite length (L). It has a starting point and an ending point:

$L = x_{end} - x_{start}$

In between these start and end points lies a certain amount of x-values that we will be analyzing. We can choose how many of these values we will evaluate, each point goes along the number scale from the start to end [0, 1, 2, 3... n]. The length divided by the number of n-values gives us our dx:

$n_{values} = [0 \ldots x_{end}]$
$dx = (x_{n-end} - x_{n-start})/n_{values}$

The basic notion is that an n-value has a corresponding x-value:

$n_{value} = x_{value}$

If each n-value represents a certain x-value we can say dx is step forwards or backwards, we can state:

$n_{value} + 1 = x_{value} + dx$
$n_{value} - 1 = x_{value} - dx$

We understood that temperature is a function of space T(x), therefore each x-value should have a corresponding T-value:

$T_{n_{value}} = T(x_{value})$
$T_{n_{value}+1} = T(x_{value} + dx)$
$T_{n_{value}-1} = T(x_{value} - dx)$

Now we must define our boundary condition for time. Similar to our spatial boundary condition we have a finite amount of time we are analyzing. We have an initial time, usually at 0, and end time:

$t_{total} = t_{initial} \rightarrow t_{end}$

Unlike how we solved for our spatial step we need to use a stability condition for our time step. A time stability condition ensures smooth diffusion over time. Solving our time coefficient:

$dt \leq 0.5(dx^2/\alpha)$

We know our time will increase by an amount dt at a constant rate until we reach the time we have chosen for the simulation to end. Therefore there is a certain amount of iterations that we need to analyze. To find the amount of iterations our simulation will run can be found by the total time divided by our time step:

$$m_{values} = t_{total}/dt$$

We have the same basic understanding that every m-value corresponds with a t-value, and will increase to a new m-value for every increasing time step:

$$m_{value} = t_{value}$$
$$m_{value} + 1 = t_{value} + dt$$

Knowing that temperature changes as time moves forward, we come to the knowledge that temperature is a function of time T(t). Each temperature value then will have a corresponding time value:

$$T^{m_{value}} = t_{value}$$
$$T^{m_{value}+1} = T(t_{value} + dt)$$

We finally need to define our thermal diffusivity term, this term remains as a coefficient that describes at what speed energy will diffuse through our material:

$$\alpha = k/\rho c_p$$

Now that we have constructed how to define our boundaries and how we can show the relationship for time and space to temperature with the term $T_n^{\ m}$, we can reconstruct our equation:

$$(T_n^{\ m+1} - T_n^{\ m})/dt = \alpha[(T_{n+1}^{\ m} - 2T_n^{\ m} + T_{n-1}^{\ m})/dx^2]$$

Since we are solving for what the change in temperature will be at a given point, we want to solve the temperature at a point n forward in time m+1. We can rearrange the equation so we can solve for our temperature change:

$$T_n^{\ m+1} - T_n^{\ m} = \alpha(dt/dx^2)[(T_{n+1}^{\ m} - 2T_n^{\ m} + T_{n-1}^{\ m})]$$

$$T_n^{\ m+1} = T_n^{\ m} + \alpha(dt/dx^2)[(T_{n+1}^{\ m} - 2T_n^{\ m} + T_{n-1}^{\ m})]$$

The boundary conditions imply the solution for $u(y,t)$ will be solved on a semi-finite domain ($y \in [0,\infty)$). Judgment suggests the solution to be dependent on time and approaches its steady state, ie. a transient solution associated with the homogenous solution. The general solution is $f(x) = h(x) + s(x)$ where $h(x)$ is the transient homogenous solution, and $s(x)$ is the steady state solution. In the case of this equation the solution can be written as $u(y,t) = h(y,t) + s(y,t)$. The steady state solution requires there to be no change in u over time, thus reducing the equation:

$$\frac{d^2u}{dy^2} = 0$$

To solve this differential equation integrate twice with respect to y:

$$u_s(y) = C_1 y + C_2$$

The coefficients of integration C are determined by the boundary conditions. Begin by applying the first boundary condition, the no slip condition where At $y = 0\ u(0,t) = 0$:

$$u_s(0) = C_1(0) + C_2$$
$$C_2 = 0$$

Followed with the second boundary condition, as y approaches infinity we have uniform flow, At $y \to \infty\ u(y,t) = U$:

$$u_s(y) = C_1 y$$

For u to remain finite and equal to U, $C_1 = 0$. Thus giving us the final solution for the steady state equation is:

$$u_s(y) = U$$

Transforming the homogenous solution, $u(y,t) = h(y,t) + s(y,t)$ into $u(y,t) = h(y,t) + U$. Solving for the homogeneous function, $h(y,t) = u(y,t) - U$ where $h(y,t)$ represents the deviations from the steady state. Substituting $h(y,t)$ back into the simplified boundary layer equation:

$$\frac{\partial h}{\partial t} = \nu \frac{\partial^2 h}{\partial y^2}$$

There exists boundary conditions for $h(y,t)$ that are in relation to those of $u(y,t)$. We have solved the boundary conditions for $u(y,t)$, inherently the solutions for boundary conditions of $h(y,t)$ would be the solution of $u(y,t)$ minus the steady state solution which is the freestream velocity U:

$$\text{At } y = 0\ u(0,t) = 0 \mid \text{therefore } h(0,t) = -U$$
$$\text{At } y \to \infty\ u(y,t) = U \mid \text{therefore } h(y,t) = 0$$
$$\text{At } t = 0 \text{ when } y > 0, u(y,0) = U \text{ and } u(0,0) = 0 \mid \text{therefore } h(y,0) = 0 \text{ and } h(0,0) = -U$$

Since $u(x,y) = g(\eta)$ then, $u'(x,y) = g'(\eta)$ and $u''(x,y) = g''(\eta)$. Expressing the partial terms of η instead of x and y

As Ω is in motion, we understand it will remain until some force acts upon it due to the inertia of Ω. Hence a particle in a moving fluid will remain in motion until some force acts on it. The net forces acting on this particle will be 0 if the velocity of the particle is constant. It is understood the acceleration of an object is directly proportional to the forces being applied and inversely proportional to the object's mass, otherwise stated as the sum of forces acting on an object is equivalent to the flux of momentum, $\Sigma F = ma$. When considering a single particle's motion the simple momentum equation is sufficient to describe the relationship between the momentum flux and the forces acting on the particle. Since the mass of the particle remains constant, for this reason any change in density through space and time must be 0, $\Delta\rho = 0$:

$$\Sigma F = m\frac{d\tilde{u}}{dt}$$

Imagine an infinite empty volume in which exists a single particle Θ. If Θ is the only known matter to exist in this volume, logically the total mass within the volume at $t = 0$ is equivalent to mass at anytime when $t > 0$,

$$m_\Theta(t_0) = m_\Theta(t_n).$$

Thus there must be no change in mass at any given time within the volume,

$$m_\Theta(t_n) - m_\Theta(t_0) = 0,$$

or,

$$m_\Theta(t_n - t_0) = 0.$$

An important conservation law that satisfies the conditions that matter cannot be created or destroyed within our system.

Imagine an infinite 3D system that is completely desolate except for a single particle. This one particle is moving solely in the x direction with no forces acting on it. That path with the least action must be in a straight line from a-b. If a force were to act on the particle the path of least action would shift. Thus any force that acts upon the particle will affect the direction in which inherently changes the action because that path a-b has changed.

Can I say that the total energy of a particle, the kinetic, internal, and potential energy minus the work being done on the particle, is equal to the new energy of the particle?

The change in work than must be proportional to how much the velocity changes,

$$dW = \frac{1}{2} m_{\Theta} du^{2}.$$

Imagine an infinite empty volume which extends in every direction, thus the volume can be defined spatially as $V = xyz$. Within this volume there exists a single particle O. It can be stated that O contains scalar properties of matter which we will call Z. If O is the only known matter to exist in this volume, logically the total amount of O within V is O. Imagine if O was moving at some constant rate u in the x-direction, regardless of the velocity of O we understand that O will remain consistent in V. Since V stretches out infinitely, analyzing a small control space, $dV = dxdydz$, which intersects the path of a moving particle can allow for further understanding about how matter must be conserved within our system. Let us say our particle enters dV at some point $a(0, y, z)$ at $t = 0$ and exits at point $a(x, y, z)$ at some time $t > 0$. We want to see the total amount of O moving through dV. To visualize the context of this situation mathematically we can integrate O along its path from a to b,

$$\int_{a}^{b} O dx.$$

O is considered to have some velocity, otherwise O would not be able to transport itself from a to b, hence we can transform our integral from one dependent of space to time using the relationship $dx = udt$ where $a \to t = 0$, $b \to t > 0$, and $u = C$,

$$u \int_{0}^{t} \Omega dt,$$

thus,

$$u(Ot - O0).$$

Since Ω is some function of time $O(t)$ and $O = O$ at all times, than $O(t2) - O(t1)$ must equal 0. If so, u could or couldn't be considered a constant when considering the case of a single particle, even if u does not remain constant as long as O contains some velocity, $|u| > 0$, as $t \to \infty$ O must exit our control space. Hence we can say that any changes or fluctuations of $O(t)$ within some control space dV must be 0 and u is also a function of time $u(t)$, and is a core factor of the conservation equation,

$$uO(t_2 - t_1) = 0.$$

Rewritten,

$$deltauO/deltat = 0.$$

Taking the limit as $\Delta t \to 0$ our equation transforms into a derivative,

$$duO/dt = 0$$

1. stl introductions and model set up *(Step 1)*
2. Creating a domain to surround my model *(step 2)*
3. Align model and domain with origin *(Step 3)*
4. Meshing the interior and making it more dense closer to the surface *(step 4)*

*****MAKE SURE TO PLOT A 2D SIDE VIEW THAT SHOWS JUST THE MODEL AND THE MESH, SHOULD BE IN THE SAME VIEW AS ALL THE PLOT*****

5. Now all data for cells(tetrahedral cell): *(step 5)*
   - Centroids G
     - Position xyz
       - An important reason for centering our model, is because we give each G a positional vector $\dot{G}$.
     - 4 Faces f
       - 4 Normals n
         - Again found using the curl and vertices and more.
       - 4 Areas A
         - The area of the face can be found by using the distance 3 of the 4 vertices of the cell.
         - Areas of the face are unique to its face.
       - 4 Neighbors N
         - Share a face with G but is also some distance and direction away so that must be found..
         - It is another centroid so it has all the properties of the centroid we are studying currently.
         - That means it contains a mass, volume, velocity, potential value and position.
         - An important note is that ghost centroids exist they are centroids that don't exist but must exist because our simulation is finite so some centroids on the boundary of the outer domain or the surface of our model dont have neighboring centroids across those faces, so we implement a ghost centroid whose values are determined by boundary conditions.
     - Velocity $\bar{u}$
       - Where $\bar{u} = \nabla\varphi$.
       - And every centroid has some velocity value which is judged by $\nabla\varphi$ of its neighbors.
     - Potential Scalar φ
       - If the gradient of a centroid is found using its neighbors then all centroids must have a potential value initially.
     - Volume V
     - Density ρ
     - Pressure P

*****NOW PLOT THE 2D VIEW OF THE MESH AND MODEL BUT NOW WITH ALL CENTROIDS(red), NORMALS(purple),AND MODEL SURFACE FACE NORMALS(blue/green). ALSO PLOT THE GHOST CENTROIDS(yellow)*****

6. Initial conditions before code is starting to approximate flow. *(step 6)*
   - All centroids initially are given a value φ based on $\varphi = Ux$ and φ_G unique position

*****PLOT ANOTHER 2D PLOT NOW JUST REAL AND GHOST CENTROIDS BUT THEIR COLOR IS DEPENDENT ON THEIR φ VALUE*****

7. Boundary conditions that must be satisfied and equations that will be used. An important note for boundary conditions is that they are enforced constantly throughout the simulation.: *(Step 7)*
   - $\bar{u} = \nabla\varphi$
     - Using a gradient scheme we can approximate using centroids
       - $\iiint(\nabla\varphi)dV = \oiint(\varphi n)dA$, or discretized, $\nabla\varphi dV = \sum\varphi n A$.
       - $\nabla\varphi_G \approx \frac{1}{dV}\sum_{f\in\partial G}\varphi_f n_f A_f$ and that velocity $\bar{u}_G \approx \frac{1}{dV}\sum_{f\in\partial G} A_f \frac{\varphi_{N_f}-\varphi_G}{||\dot{N}_f-\dot{G}||}\dot{d}_{GN_f}$
       - But for our case $\nabla\varphi_G \approx \sum_{f\in\partial G}\frac{\varphi_{N_f}-\varphi_G}{||\dot{N}_f-\dot{G}||}\dot{d}_{GN_f} \approx \bar{u}_G$
   - $\iiint_V(\nabla\cdot\bar{u})dV = \oiint_{\partial V}(\bar{u}\cdot n)dA = 0$
     - This must be proved using all the centroids that are on ∂V and calculate the flux between that centroid and its neighbors, one of which will be a ghost neighbor whose initial constant conditions are $\varphi = Ux$.
       - $\sum_{f\in\partial V}\frac{\varphi_{N_f}-\varphi_G}{||\dot{N}_f-\dot{G}||}(\dot{d}_{GN_f}\cdot n_f)A_f = 0$ where N_{fV} 'ghost centroids' always abide to $\varphi = Ux$. basically like inlet outlet flow but around the whole thing
   - Fluid should never flow into or out of my model, thus we can say that on the surface S, so $\bar{u}\cdot n_S = 0$
     - This is key and also flow at the surface of the model must always be tangential to it.
     - $\sum_{f\in\partial G}\frac{\varphi_{N_{fS}}-\varphi_G}{||\dot{N}_{fS}-\dot{G}||}(\dot{d}_{GN_f}\cdot n_S) = 0$

- It must be stated N_{fS} and $\dot{N}_{fS}$ in this case they are 'ghost centroids' and their value must satisfy the boundary condition, in this case $\bar{u} \cdot n_S = 0$.

❖ $\iiint_G (\nabla \cdot \bar{u}) dV = \oiint_{\partial G} (\bar{u} \cdot n) dA = 0$

➢ Since the flux and gradient both are variable with the face, normal vector, area and distance then we can say the volume they will be approximated through is the same and therefore use the same faces and properties.

- $\sum_{f \in \partial G_k} \frac{\varphi_{N_f} - \varphi_G}{||\dot{N}_f - \dot{G}||} (\dot{d}_{GN_f} \cdot n_f) A_f \approx \oiint_{\partial G} (\bar{u} \cdot n) dA = 0$
- $\Phi_f \approx \frac{\varphi_{N_f} - \varphi_G}{||\dot{N}_f - \dot{G}||} (\dot{d}_{GN_f} \cdot n_f) A_f$

➢ The sum of fluxes through each cell must remain 0, this is the laplacian where $\nabla^2 \varphi = 0$. It can be solved using an $AX = B$ matrix where its $X = \varphi$.

➢ The goal being to achieve solution $\varphi = \frac{B}{A}$

- $\sum_{f \in \partial G_k} \frac{(\dot{d}_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||} \varphi_{N_f} - \varphi_G \sum_{f \in \partial G_k} \frac{(\dot{d}_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||} = 0$
- $\varphi_G = \frac{\sum_{f \in \partial G_k} \frac{(\dot{d}_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||} \varphi_{N_f}}{\sum_{f \in \partial G_k} \frac{(\dot{d}_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||}}$ is the solution to $A\varphi_G = B$

❖ We have our governing equations to find a solution for φ_G. We can group terms to simplify a little further.

➢ We can say $\frac{(\dot{d}_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||}$ is equivalent to the weight of the matrix, it's what affects the relationship between a centroid and its neighbors.

- $w_f = \frac{(\dot{d}_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||}$

➢ Thus we can reduce the matrix equation $A\phi = B$ to the form $\sum w_f \varphi_G = \sum w_f \varphi_{N_f}$

- $\varphi_G = \frac{\sum w_f \varphi_{N_f}}{\sum w_f}$

8. Finally boundary conditions are applied to the surface of the model and then a gradient must be found, the computer must find all φ values to satisfy the continuity conditions and boundary conditions. *(Step 8)*

❖ To solve iteratively we must say before it is updated by its neighbors its at some initial state k and after it is updated it is at some state k.

➢ $\varphi_G^{(k+1)} = \frac{\sum w_f \varphi_{N_f}^{k}}{\sum w_f}$ but if the neighbor has been updated already $\varphi_{N_f}^{(k+1)}$ we can use the neighbors updates version if not we will remain with the original version $\varphi_{N_f}^{k}$.

- $\varphi_G^{(k+1)} = \frac{\sum w_f \varphi_{N_f}^{(*)}}{\sum w_f}$ where * is to be determined depending on whether the neighbors have or have not been updated.
- After each step forward it is considered that it has been updated and that is now its new initial value k

❖ We want the system to converge to some point, we don't want it to run forever.

➢ $|\varphi_G^{(k+1)} - \varphi_G^{(k)}| \leq \varepsilon$

- Where ε is some error term, $\varepsilon \approx 10^{-2} \rightarrow 10^{-6}$, that once the simulation runs to it stops iterating for that cell

➢ We can say a residual is left before the iteration, $\sum w_f \varphi_{N_f}^{(*)} - \varphi_G^{k} = \Re_G$

- $R = \sqrt{\sum_G \Re_G^2}$ and once R has converged to ε then we know to end the simulation.

*****PLOT THE FINAL PLOT IN THE SAME 2D VIEW AND NOW JUST PLOT ONLY CENTROIDS BUT THEIR COLOR IS DEPENDENT ON THEIR VELOCITY, USE THE TURBO COLOR GRADIENT FOR THIS*****

- After all values are initialized from step 6 we need to begin solving our flow.

- For centroids on the outer boundary, i.e. shares a face with the outer boundary than their phi values must be fixed after initializing meaning for G on the surface of the outer domain their phi values are always determined by $\varphi = Ux$. So for these centroids their phi values can never change.
- Now for centroids on the surface of the model we must say that the $\bar{u} \cdot n = 0$. This means that the change in phi through that face = 0, so that needs to be maintained for the centroids on the surface. So the gradient through one of their 4 faces that is on the model surface S must always 0.
- We also know that flux must be conserved throughout the entire system $\iiint_V (\nabla \cdot \bar{u}) dV = \oiint_{\partial V} (\bar{u} \cdot n) dA = 0$. we can discretize the equation and say this has to be conserved through each cell. $\iiint_G (\nabla \cdot \bar{u}) dV = \oiint_{\partial G} (\bar{u} \cdot n) dA = 0$. thus the discretized equation for each cell is, $\sum_{f \in \partial G_k} \frac{\varphi_{N_f} - \varphi_G}{||\dot{N}_f - \dot{G}||} (đ_{GN_f} \cdot n_f) A_f \approx \oiint_{\partial G} (\bar{u} \cdot n) dA = 0$. This must be true for every centroid in V.
- We can solve for each centroid using $AX = B$ matrix where its $X = \varphi$. The goal being to achieve solution $\varphi = A^{-1}B$. $\sum_{f \in \partial G_k} \frac{(đ_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||} \varphi_{N_f} - \varphi_G \sum_{f \in \partial G_k} \frac{(đ_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||} = 0$. $\varphi_G = \frac{\sum_{f \in \partial G_k} \frac{(đ_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||} \varphi_{N_f}}{\sum_{f \in \partial G_k} \frac{(đ_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||}}$ is the solution to $A\varphi_G = B$.
- Now we have to maintain boundary conditions, we know the outer centroids values are fixed so their values never change throughout the iteration. They will constantly be affecting centroids that share faces with them.
- Next boundary condition is that $\bar{u} \cdot n = 0$. So for centroids that have a face that is on S then that face would have 0 changes in phi. So a centroid on the surface has four faces, 3 of the faces have neighbors, one lies on the surface so no neighbor but we know that change of phi through that face = 0. So that is a fixed value. So the equation for centroids would be $\varphi_{G_S} = \frac{\sum_{f \in \partial G_s} \frac{(đ_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||} \varphi_{N_f} + F_s}{\sum_{f \in \partial G_s} \frac{(đ_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||}}$ where F_S is the flux through the face on the surface, and that always equals 0. Since that face doesn't have a neighbor across from it the change in phi is fixed to 0. The gradient through F_S is always 0.
- Now that we know what equations to use to solve for our flow with boundary conditions we can begin solving using an iterative method. To solve iteratively we must say before it is updated by its neighbors its at some initial state k and after it is updated it is at some state k.
  - ➢ $\varphi_G^{(k+1)} = \frac{\sum w_f \varphi_{N_f}^{k}}{\sum w_f}$ but if the neighbor has been updated already $\varphi_{N_f}^{(k+1)}$ we can use the neighbors updates version if not we will remain with the original version $\varphi_{N_f}^{k}$.
    - $\varphi_G^{(k+1)} = \frac{\sum w_f \varphi_{N_f}^{(*)}}{\sum w_f}$ where * is to be determined depending on whether the neighbors have or have not been updated.
    - After each step forward it is considered that it has been updated and that is now its new initial value k
    - So surface centroids it has to be noted that this must always remain true during each iteration, $\varphi_{G_S} = \frac{\sum_{f \in \partial G_s} \frac{(đ_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||} \varphi_{N_f} + F_s}{\sum_{f \in \partial G_s} \frac{(đ_{GN_f} \cdot n_f) A_f}{||\dot{N}_f - \dot{G}||}}$ where F_S is the flux through the face on the surface, and that always equals 0 til convergence and at each step of iteration.
    - And for centroids on the outer boundary have fixed values for the whole time til convergence.
  - ❖ We want the system to converge to some point, we don't want it to run forever.
    - ➢ $|\varphi_G^{(k+1)} - \varphi_G^{(k)}| \leq \varepsilon$
      - Where ε is some error term, $\varepsilon \approx 10^{-2} \rightarrow 10^{-6}$, that once the simulation runs to it stops iterating for that cell
    - ➢ We can say a residual is left before the iteration, $\sum w_f \varphi_{N_f}^{(*)} - \varphi_G^{k} = \text{ϟ}_G$
    - ➢ Once the residual matches the convergence limit or if the amount of iterations runs out this update part of the simulation stops.
  - The iterations will begin where phi values are the lowest, so where flow enters, and then it will update down stream each time. This is key.
  - Once this has been converged we will use the least squares gradient model to find our velocity values and plot them on plot 4

The airfoil body, B, we are choosing to simulate over must be some bounded volume, $B \subset \mathbb{R}^3$. The surface of the airfoil S is the boundary of the airfoils body, ∂B. If ζ is some larger body that encases B, let $\partial\zeta$ be the boundary that defines this outer domain. The space inbetween S and $\partial\zeta$ will be the control volume where flow analysis will occur which we will call V.

This set control volume gives a manageable space for discretization; say V is a composition of tetrahedral cell ς containing some little volume dV where $\varsigma_i \cap \varsigma_j = \emptyset$ for $i \neq j$,

$$V = \bigcup_{G\in} \varsigma_k.$$

python3 ~/PycharmProjects/potentialflowcomplexgeometries/"Finite Volume Analysis.py"